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A review on functionalization strategies of graphene oxide hydrogels for uranium extraction and machine learning insights

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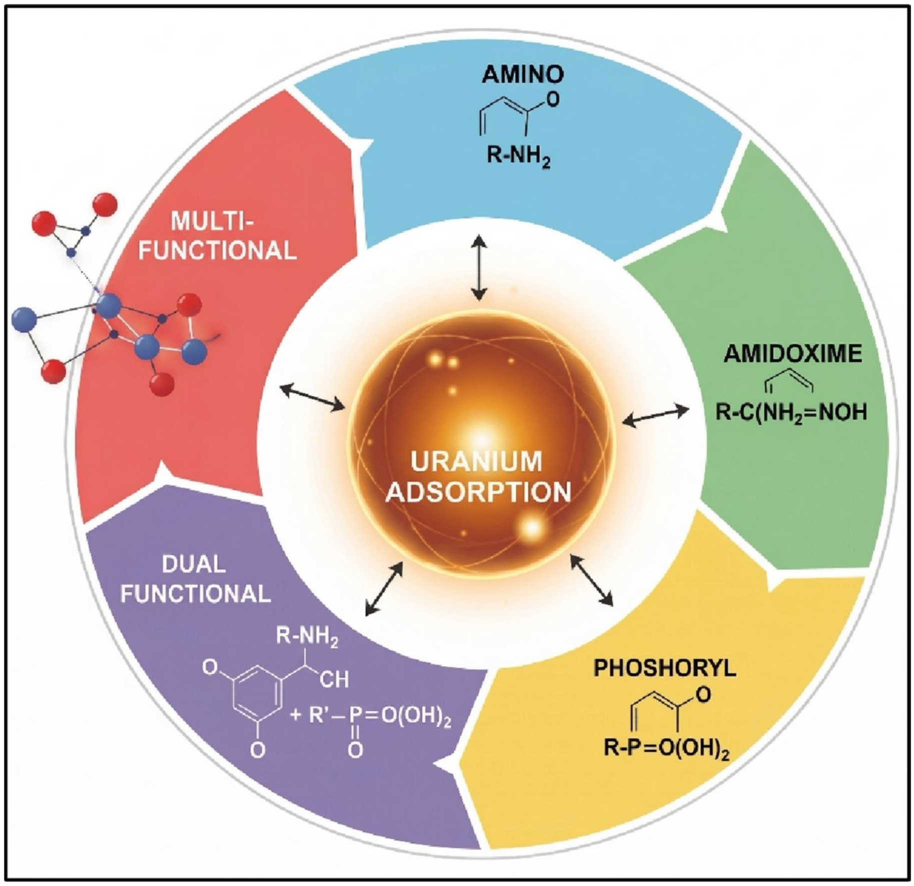
Abstract

The sustainable use of nuclear fuel and the cleanup of contaminated water sources necessitate the efficient recovery of uranium from aqueous sources. Conventional techniques for uranium recovery exhibit poor selectivity, expensive processes, and inadequate performance in complex ionic solutions such as seawater and nuclear waste with the uranium concentration of (~ 3.3ppb) and (~ 10ppm) respectively. Hydrogel adsorbents based on Graphene oxide (GO), due to their high surface area, good stability, porous three-dimensional structures, and tuneable surface chemistry, have emerged as a new class of adsorbents for uranium recovery in recent years. This review offers a comprehensive and critical review of graphene oxide-based hydrogels for the extraction of uranium (VI) from aqueous environments. A discussion of the underlying chemistry of uranium speciation, adsorption, and hydrogel synthesis identifies the structure property relationships that shows uranium capture in hydrogels. The adsorption capacity, selectivity, kinetics, and regeneration of several functionalized Graphene oxide (GO) hydrogel systems are reviewed, including reduced Graphene oxide (rGO), amidoxime functionalized, phosphoryl functionalized, and multi ligand-based systems. Substantial developments have been made, but challenges remain in the long-term stability, scalability, antifouling properties, and practical deployment of these systems. By critically comparing various functionalization schemes and identifying knowledge gaps, this review aims to map a course for future research on multi-ligand system GO based hydrogels and integrate machine learning (ML) approaches to predict adsorbent design, prioritize functionalization strategies, and reducing experimental trial-and-error. This paper offers a data-driven methodology for the logical design of next-generation GO adsorbents for uranium removal, providing useful information for nuclear wastewater treatment and environmental remediation.

Keywords Uranium, Graphene oxide (GO), Reduced graphene oxide (rGO), Amidoxime, Phosphoryl, Amino, Hydrogels, Machine learning (ML)

Graphical Abstract



1 Introduction

Uranium's radioactive properties and natural chemical reactivity make it a significant risk to both aquatic environments and human health, needing necessary remediation measures to limit its effects [1]. Uranium is seen as uranyl cations (UO_2^{2+}) in seawater and is present at a dilute concentration of ~ 3.3 ppb, which are extremely soluble and mobile in water and have a strong tendency to form coordination complexes with dissolved species and coexisting ligands. These complexes have an impact on the transport and bioavailability of these substances [2]. Uranium removal from sea water and industrial/waste effluents (~ 10ppm) are therefore an important to public health and environmental protection measure [3]. Although, extracting uranium from seawater is highly challenging due to several factors. The concentration of uranium is extremely low (~ 3.3 μg/L), while the salinity is very high, with metal ion concentrations ranging from 0.6 to 0.7 mol/L. Additionally, uranium exists in multiple complex forms such as UO_2^{2+}, $UO_2(CO_3)_2^{2-}$, and $UO_2(CO_3)_3^{4-}$. The process is further complicated by environmental conditions, including slightly alkaline pH (~ 8), fluctuating temperatures (12–24 °C), the presence of abundant competing ions like Na^+, Mg^{2+}, Cl^-, and CO_3^{2-}, as well as diverse biological materials [4].Uranium is often extracted from water using adsorption, ion exchange, chemical precipitation, solvent extraction and membrane filtration, which either bind dissolved uranyl ions or change them into a different solid form. Uranium ions bind to the surfaces of solid materials such as hydrogels, aerogels, covalent organic frameworks (COFs), and other porous adsorbents during adsorption. Functional groups such as hydroxyl, carboxyl, phosphate, and amidoxime are present in these materials and form stable inner-sphere complexes with uranium (U(VI)). Uranium recovery is made possible by the material's high adsorption capacity and ability to regenerate [5–10]. Ion exchange uses solid materials with fixed charged groups. These substances exchange their initial ions for uranyl ions found in water. Strong anion exchange resins and specially designed metal-organic frameworks can substitute negatively charged uranium complexes for their existing anions. Even in complex water environments and at extremely low uranium concentrations, this approach is extremely successful and selective. Acids or other suitable solutions can subsequently be used to recover the extracted uranium [11–16]. Chemical precipitation removes uranium by transforming it to an insoluble form that can be easily extracted from water. Uranyl ions form complexes before being trapped within iron hydroxide or oxide particles in processes like iron electrocoagulation with organic ligands. Up to 99% efficiency is possible when these particles settled. Because it concentrates uranium into a smaller and manageable volume, this technique is very helpful for treating contaminated wastewater and mining tailings [17–19]. Among the above discussed uranium extraction techniques demonstrated in the Fig. 1, adsorption using advanced sorbents is generally considered the most effective due to its high selectivity, large adsorption capacity, fast kinetics, and good renderability, even in complex matrices [20–22]. Compared with precipitation and conventional ion exchange, functionalized adsorbents such as ion-imprinted polymers, chelating resins, and graphene-based materials reduce sludge generation and secondary waste while maintaining efficiency across wider pH ranges [23–25]. However, adsorption can face limitations related to sorbent cost, fouling, and performance loss over multiple cycles, necessitating optimization for large-scale deployment [26, 27].

These techniques frequently have limitations such as low removal efficiency at trace concentrations, high operating and maintenance costs, and difficulties scaling up for large-volume treatment [28]. These disadvantages have led researchers to concentrate

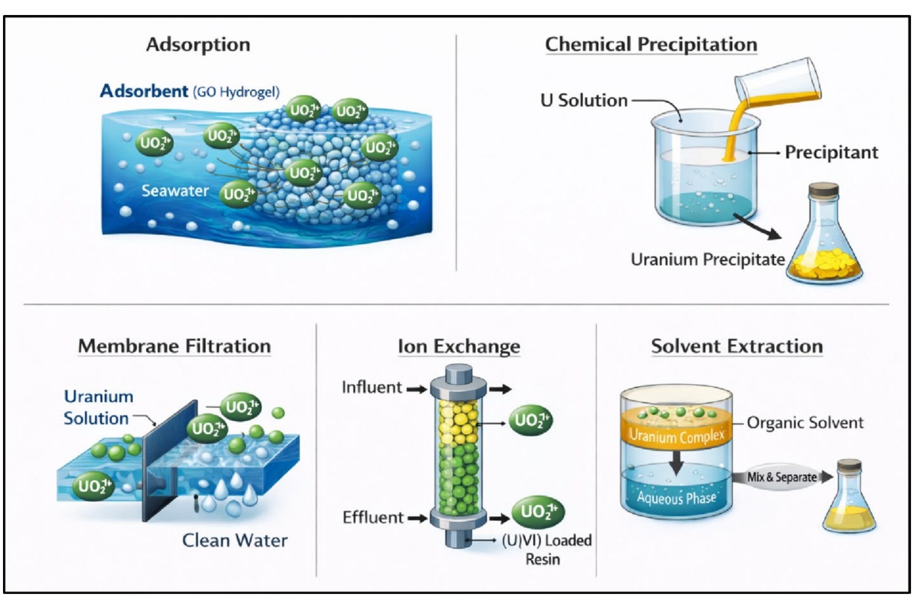


Fig. 1 Uranium extraction methods

more on creating new, reusable, and effective adsorbent materials that can efficiently extract uranium from aqueous solutions [29]. Because of their large specific surface area, high surface functionality, and tunable chemistry that promote robust inner-sphere complexation with uranyl ions, carbon-based materials and its derivatives have attracted a lot of interest among researchers [30]. Also, the increasing complexity of adsorption systems, as well as the large number of influencing elements such as pH, surface chemistry, and porosity, make it challenging to optimize materials using only traditional methods. In this regard, ML has become a promising method for predicting uranium uptake performance across various carbon-based materials, analysing adsorption behavior, and identifying important controlling elements [31].

The goal of this review is to identify the best carbon-based materials for uranium extraction. It shows how various functional groups affect adsorption performance and tackles important issues including stability, scalability, and practical use. It also emphasizes ML as a contemporary data-driven method to direct the development and improvement of advanced adsorbents, allowing for more effective and sustainable uranium extraction.

2 Carbon materials for uranium extraction

Carbon-based absorbents have emerged as a highly promising materials for uranium extraction from aqueous environments due to their variable surface chemistry, large specific surface area, and excellent chemical/thermal stability [32]. Carbon- based materials such as activated carbons, GO, and carbon nanotubes can be readily functionalized with ligands including oxygen, nitrogen, or sulfur, greatly increasing their affinity and selectivity toward UO_2^{2+}. These materials are able to adsorb the UO_2^{2+} ions even at small quantities in seawater and nuclear effluent [33, 34]. Also, their strong carbon structure allows for good regeneration and reusability, and their hierarchical porosity promotes fast mass transfer and high adsorption capabilities, making them appealing for real-world uranium recovery applications [35].

2.1 Activated carbon

Activated carbon is often used to extract uranium from water due to its large surface area and porous structure, which helps in adsorption [36]. However, pure activated carbon's ability to selectively collect uranium is limited, especially in complex water systems, because it lacks enough binding sites [37]. Researchers have changed activated carbon and mixed it with other materials to increase the number of active sites for uranium binding in order to enhance its effectiveness [38]. Combining it with GO, which provides oxygen-containing functional groups that improve uranium adsorption through complexation and ion exchange is one efficient method [39].Chen et al. showed that a GO-activated carbon felt composite achieved a much greater adsorption capacity of 298 mg/g at pH 5.5 as compared to 173 mg/g for pure activated carbon, with fast adsorption which reached equilibrium in 30 min and corresponded to the Langmuir model [40]. Overall, these composites increase structural stability and adsorption efficiency, making them appropriate for real-world uranium extraction applications.

2.2 Graphene oxide and functionalized graphene derivatives

GO is one of the most extensively studied materials for uranium extraction because it includes a large number of oxygen-containing functional groups such as hydroxyl, epoxy, and carboxyl groups, which provide active sites for uranyl ion binding [41]. However, surface modification can improve the adsorption capacity of pristine GO by increasing the number of effective binding sites [42]. For example, since carboxylated GO (GO–COOH) has a larger density of carboxyl groups, which improve chelation and electrostatic interactions with U(VI) ions, it exhibits better uranium adsorption than unmodified GO [43]. rGO has also received attention for its superior structural stability and adsorption capability when compared to GO. Adsorption effectiveness is increased by rGO-based porous composites' enhanced diffusion pathways and increased surface area [44]. For example, a rGO/ZIF-67 aerogel composite exceeded many traditional adsorbents with a remarkably high uranium adsorption capacity of 1888.55 mg/g at an ideal pH of 4.01. It was discovered that the adsorption process was endothermic and spontaneous, with pseudo-second-order kinetics indicating chemisorption. Also, after five regeneration cycles, the material maintained roughly 73% of its adsorption capacity and demonstrated strong selectivity toward uranium in the presence of competing ions [28].

Overall, GO and its derivatives are among the most promising carbon-based materials for uranium recovery applications because of their high adsorption capacity, superior selectivity, and structural stability.

2.3 CNTs (carbon nanotubes) and CNT composites

Carbon nanotubes (CNTs) are useful carbon-based materials for uranium adsorption due to their tubular structure, high mechanical strength, and huge surface area [45]. Pristine CNTs have low affinity for uranium and typically need surface oxidation or functionalization to provide active binding sites [46]. A maximum adsorption capacity of 333.13 mg/g at pH 6.0 was reached by Wang et al.'s investigation of uranium adsorption on GO and CNT-based materials. The Langmuir isotherm and pseudo-second-order kinetics of the adsorption indicate that monolayer chemisorption is mostly caused by uranyl ions and nitrogen-containing functional groups. Additionally, the material demonstrated great selectivity toward uranium in the presence of competing metal ions and retained over 87.2% of its adsorption capacity after five cycles, showing good reusability [30].

To address the aggregation concerns associated with CNTs, CNT-based Three-dimensional (3D) porous carbon networks have been developed [47]. In this case, at pH 5.0, a porous aerogel made with chitosan and carboxylated carbon nanotubes had a maximum adsorption capacity of 307.5 mg/g. Langmuir and pseudo-second-order models were used for the adsorption, and uranium was mostly bound by functional groups that contained oxygen and nitrogen, such as amine and carboxyl groups. Also, the aerogel's porous structure boosted adsorption efficiency and mass transfer [48].

2.4 3D porous carbon networks

Carbon aerogels are a specialized class of three-dimensional porous carbon materials with high porosity, low density, and interconnected networks. They offer a large number of accessible adsorption sites and speed up mass transfer for uranium removal. Among these, GO-based aerogels have been extensively researched because of their stable three-dimensional structure and abundance of oxygen-containing functional groups, which improve uranium binding through surface complexation and ion exchange mechanisms. For example, according to Langmuir isotherm and pseudo-second-order kinetic models, a GO-chitosan composite aerogel showed excellent uranium adsorption ability with a maximum capacity of 929.16 mg/g, suggesting monolayer chemisorption. The combined effect of porous structure and numerous functional groups, which enhance both accessibility and binding strength for U(VI) ions, was attributed with the high efficiency [29].

In contrast, carbon-based hydrogels have evolved as flexible 3D networks with high water content and variable porosity, providing superior diffusion paths and faster adsorption kinetics. A UiO-66/calcium alginate/hydrothermal carbon composite hydrogel achieved a maximum uranium adsorption capacity of 337.77 mg/g and removal rates exceeding 98% at 3–8 pH [49]. Aerogels often offer greater structural stiffness and surface area, whereas hydrogels offer greater flexibility and stronger interaction with aqueous fluids. Overall, hydrogels and aerogels are both promising 3D porous carbon structures for uranium extraction. Hydrogels offer better mass transfer and operational flexibility, while aerogels offer higher adsorption capacity [29, 50].

Table 1.shows that GO-based aerogel systems dominate other carbon-based materials in terms of adsorption performance (up to 1888.55 mg/g and ~99% efficiency). This is due to their high surface area and large number of functional groups. However, scalability and regeneration are still major drawbacks. While functionalized GO (such as GO-COOH) improves chelation and adsorption efficiency, activated carbon composites offer strong stability but poor selectivity. Because of their porous structures and large number of accessible binding sites, 3D GO-based aerogels outperform CNT-based systems that need further modification in terms of adsorption capabilities. Therefore, GO and GO-based aerogels and hydrogels are thought to be the best carbon materials for high-efficiency uranium extraction, particularly in diluted environments like seawater.

3 Structure and adsorption capability of GO

Graphite is chemically oxidized to create GO, a two-dimensional substance [51]. Due to oxygen containing groups like hydroxyl, carboxyl, and epoxy, GO, a typical 2D material, is made up of single layers of carbon atoms arranged in a hexagonal lattice being

Table 1 Comparative analysis of carbon-based materials for U(VI) removal

Material	Adsorption capacity (mg. g^{-1})	Optimal pH	Isotherm and Kinetic Model	Limitations	Pore size (nm)	Surface Area (m2/g)	Refs.
GO-Activated Carbon Felt Composite	298	5.5	Langmuir / Pseudo-second-order	Moderate selectivity; affected by competing ions	0.71	–	[40]
rGO/ZIF-67 Aerogel	1888.55	4.01	Langmuir / Pseudo-second-order	Complex synthesis; cost; possible leaching	0.7–100+	962	[28]
GO/GO Nanoribbons/Alginate Aerogel	929.16	~6	Langmuir / Pseudo-second-order	Scale-up and regeneration issues	2.0–5.0	2630.0	[29]
TETA-Functionalized Carbon Nanohorns	333.13	6	Langmuir / Pseudo-second-order	High cost; lower surface area	13.2–17.3	–	[30]
Chitosan/CNT Aerogel	307.5	5	Langmuir / Rapid equilibrium	Lower capacity; pH sensitivity	–	73.2–106.5	[48]

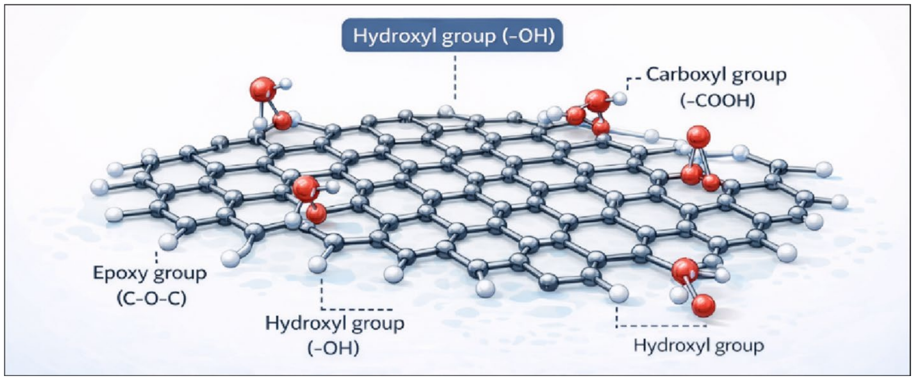


Fig. 2 Structure of GO

covered with its edge and basal plane [52]. The structure of GO is shown in Fig. 2 ., where hydroxyl, carboxyl, and epoxy are illustrated. Because these functional groups provide active sites that can bind various metal ions, GO is a promising adsorbent [53]. GO sheets are extremely dispersible in water because of these hydrophilic groups, which also increase the accessible surface area for adsorption and inhibit aggregation [54]. GO's large surface area allows for numerous binding interactions with metal ions such as uranium, improving adsorption efficiency [55] .

According to studies, GO can adsorb uranium ions from aqueous solutions by forming hydrogen bonds, combining with surface functional groups, and via electrostatic interactions [56]. Additionally, rGO, which is created by removing specific oxygen groups, might enhance adsorption by increasing π π interactions between the graphene plane and metal ions [57]. GO's variable surface chemistry provides opportunities to enhance its adsorption capability by modifying or functionalizing the material [58].In order to enhance mass transfer and provide more accessible binding sites, hydrogels and other three-dimensional (3D) configurations of graphene sheets create interconnected networks [59]. These three-dimensional structures also improve mechanical stability, allowing for repeated use in adsorption processes [60].

Graphene based hydrogels have been shown to effectively extract uranium from aqueous solutions due to their high surface area, accessible functional groups, and porous structure [61]. Because of their mechanical stability, which ensures that they retain their structure during adsorption, these hydrogels are suitable for real world applications [62]. Functionalization of GO with specific chemical groups, such as amidoxime, phosphate, or amino groups, can increase uranium adsorption by promoting stronger chemical interactions with uranyl ions [63]. Amidoxime groups in particular show exceptional selectivity for uranium because bidentate chelation stabilizes the binding [64]. Polymeric amidoxime adsorbents have been used to study sea water uranium recovery, demonstrating that this functional group is remarkably effective in extracting uranium even in the presence of other competing ions. Studies further support amidoxime's superior uranium adsorption and selectivity [65, 66]. Additionally, phosphate functionalized GO has a strong contact with uranium ions due to surface complexation, which leads to high removal efficiency [67]. Phosphoryl groups on carbon materials have also been discovered to enhance uranium adsorption and enable adsorbent recycling [68]. Also, phosphonate modified GO has shown excellent uranium capture efficiency in aqueous solutions [69]. Amino functionalized GO increases uranium's binding sites by coordinating with nitrogen atoms, increasing adsorption capacity [70]. Nitrogen doped GO has shown enhanced adsorption capability due to its ability to interact with uranyl ions [71].

3.1 Advantages of GO hydrogels

In uranium extraction, GO hydrogels are more advantageous than GO particles [72]. The 3D network prevents GO sheets from aggregating, which can reduce adsorption effectiveness in powdered forms. Hydrogels' porous shape facilitates faster adsorption by improving uranium ion mass transfer [73]. Additionally, because of its remarkable mechanical strength, the GO hydrogel can withstand multiple cycles of adsorption and desorption without degrading [74]. GO hydrogels have a higher density of active sites and a wider surface area than other carbon-based materials, which increases their overall adsorption capacity [75]. Because of their macroscopic shape, hydrogels are easier to separate from water after treatment, which reduces operational challenges [76]. Go hydrogels' structure and functional groups have allowed them to achieve greater efficiency [77]. For uranium removal, rGO hydrogels provide a dependable and reusable framework while maintaining a high adsorption capacity [78]. Because of their design versatility, 3D GO hydrogels can incorporate various functional groups that target specific metal ions [79]. Also, these hydrogels can perform effectively in a range of environmental circumstances, including different ionic strengths and pH levels [80].

3.2 Limitations of GO hydrogels

GO is widely recognized as a promising material for uranium extraction due to its huge specific surface area and presence of oxygen-containing functional groups; but it has many scientific and practical limitations [81]. Because pristine GO lacks specialized binding sites to efficiently separate uranyl ions from competing ions like Na^+, Mg^{2+}, and Ca^{2+} in complicated aqueous environments, its low intrinsic selectivity is a major disadvantage [82]. Also, GO has poor aqueous stability because nanosheets have a tendency to aggregate and form colloids, which lowers the effective adsorption capacity. When compared to functionalized derivatives, its adsorption capability is also comparatively limited. For example, pure GO exhibits modest adsorption, while treatment with ligands such as poly(amidoxime) can increase capacity to $> 1.83 \times 10^2$ mg g^{-1} [83]. Furthermore, GO's tiny particle size nature makes separation and recovery difficult, frequently requiring incorporation into magnetic composites or three-dimensional frameworks to prevent secondary pollution [84]. Lastly, the high costs and environmental issues, especially the formation of acidic waste from traditional synthesis techniques like Hummers' process, limit large-scale manufacturing [85].

3.3 Adsorption based uranium removal technologies

The adsorption technique is a feasible choice for uranium extraction due to its effectiveness and affordability [86]. GO, GO hydrogels and other carbon-based polymers are adsorbents for binding uranium ions on their surface [87]. Adsorption mechanisms were categorized into surface complexation, π-π interaction, hydrogen bonding and electrostatic attractions [88]. Kinetic approaches, such as the pseudo second order model, are often used in assessing performance and comprehending adsorption kinetics [89]. By characterizing the interaction between uranium ions and the adsorbent surface, Freundlich and Langmuir isotherm models aid in determining the maximal adsorption capacity [90]. Multiple studies have shown that uranium is absorbed by GO based materials by chemisorption processes, where uranyl ions form stable complexes with nitrogen and oxygen groups [91]. The effectiveness of adsorption is influenced by pH, temperature, and coexisting ions [92]. For example, the pH range between slightly acidic and neutral is typically optimal for uranium adsorption [93]. In multi-ion systems, selectivity becomes critical, and functionalized GO hydrogels have demonstrated the ability to remove uranium more efficiently than other competing metal ions [94]. Surface complexation modelling and thermodynamic investigations, which provide insights into the adsorption process, are used to assist the design of high-performance adsorbents [95].

Graphene based 3D hydrogels perform better in uranium extraction from aqueous solutions than traditional 2D GO sheets [96]. By combining high surface area, accessible active sites, and structural stability, these materials achieve higher capacities and faster adsorption kinetics [97]. Functionalized hydrogels also make it possible to recycle uranium from treated water and reuse the adsorbent multiple times [98]. Reusability experiments have shown that GO hydrogels retain a significant portion of their adsorption capability even after several adsorption desorption cycles [99].

4 Functionalization strategies for enhanced uranium adsorption

Functionalizing graphene-based adsorbents significantly increases adsorption efficiency and selectivity [100]. Amidoxime functional groups are often used because to their strong bonding interaction with uranyl ions, which enhances uptake from both freshwater and marine sources [101]. Uranium adsorption is increased by strong surface complexation brought on by phosphate and phosphonate modifications [102]. Amino and nitrogen containing groups can bind uranium by coordinating with nitrogen atoms, enhancing adsorption in multi-ion systems [103]. Polyamine modifications enhance uranium uptake by providing many binding sites per molecule [104]. Studies has shown that combining multiple functional groups can have a synergistic effect on uranium adsorption [105]. For example, GO materials functionalized with oxygen and nitrogen have a higher adsorption capacity than single functionalized adsorbents due to cooperative binding mechanisms [106]. By increasing the accessibility of functional groups, 3D hydrogels enable efficient interaction with uranium ions in solution [107]. Competitive adsorption tests have shown that functionalized GO hydrogels may selectively adsorb uranium even in the presence of other metal ions as Ca^{2+}, Mg^{2+}, and Fe^{3+} [108]. Thermodynamic and kinetic studies aid in the optimization of functionalization methods by offering insights into the adsorption process under real world circumstances [99]. High uranium removal efficiency, mechanical stability, and reusable qualities have been seen in graphene-based hydrogels functionalized with amidoxime, phosphate, and amino groups [109]. Because of the synergistic effects of many functional groups, these hydrogels can be employed successfully in complex aqueous environments providing a viable alternative for uranium removal [110]. Carbon-based 3D structures such as hydrogels and aerogels continue to be promising platforms for uranium removal due to their large surface area, porous structure, and tunable surface chemistry [111]. The combination of 3D hydrogel design and functionalization techniques enables the creation of high performance, selective, and reusable uranium adsorbents [112].

GO hydrogels can be the promising material for uranium removal from aqueous environments because of their structure, advantages, adsorption-based methods and functionalization techniques. Research on functionalization and 3D structural optimization will continue to improve adsorption efficiency, selectivity, and practical application in environmental remediation.

4.1 Amidoxime functional groups

Amidoxime (–C(NH_2) = NOH) is widely acknowledged as one of the most effective ligands for uranium adsorption, because it may form bidentate coordination complexes with the uranyl ion (UO_2^{2+}) [51]. The predominant form of uranium in aqueous solutions is UO_2^{2+}, a hard Lewis acid with a significant attraction for hard Lewis's base donor atoms like nitrogen and oxygen. Because amidoxime has two donors one nitrogen from the imine and one oxygen from the oxime it can form stable five membered chelate rings with uranyl ions, which leads to strong and highly selective binding [52]. GO has many oxygen containing groups (hydroxyl, carboxyl, and epoxy) on both basal planes and edges, it offers a flexible substrate for amidoxime functionalization. By acting as anchors for amidoxime ligands, these groups increase the density of active sites and make them more accessible for U(VI) adsorption. Amidoxime functionalization of GO improves selectivity in the presence of competing ions in addition to increasing adsorption capacity [55].

4.1.1 Graphene based amidoxime adsorbents

Amidoxime functionalized materials can now perform even better thanks to the development of three-dimensional (3D) graphene-based adsorbents. In their analysis of 3D graphene structures, highlights that hydrogels and macro assemblies can increase the accessibility of functional groups for adsorption, enhance mass transfer, and inhibit sheet aggregation. Because it enables more accessible binding sites and quicker adsorption kinetics, this three-dimensional structure is especially beneficial for amidoxime functionalized materials [62].

The additional research on amidoxime functionalized GO highlighted the significance of ligand density and dispersion in affecting performance. Uniform functionalization ensures that U(VI) can interact with the maximum number of amidoxime sites, whereas excessive ligand grafting can cause steric hindrance and reduce the total adsorption effectiveness. The impact of environmental elements that can change adsorption behavior, such as pH, ionic strength, and the presence of competing ions, was also highlighted in the study [63]. A GO/amidoxime hydrogel created especially for uranium adsorption was described by Wang et al. In this system, amidoxime groups were covalently bonded to the surface of GO sheets that were crosslinked to form a three-dimensional hydrogel. By preventing GO sheets from restacking, the hydrogel design improved amidoxime site accessibility and increased uranium uptake. The hydrogel has a far greater adsorption capacity than both pure GO and polymeric amidoxime adsorbents, according to adsorption studies. Inner sphere complexation was verified by spectroscopic investigation, where uranyl ions coordinated directly to the amidoxime groups' nitrogen and oxygen atoms. Additionally, the hydrogel demonstrated good reusability, continuing to function after several cycles of adsorption and desorption [64].

4.1.2 Polymeric amidoxime adsorbents

Polymeric amidoxime adsorbents have been thoroughly investigated for uranium recovery, especially from saltwater, in addition to graphene-based systems. Polymeric amidoxime materials were shown by Seko et al. to be able to selectively adsorb U(VI) even in the presence of large quantities of competing cations such Na^{+}, Mg^{2+}, and Ca^{2+}. The amidoxime group and uranyl ion chelated to form stable complexes throughout the adsorption process, and the adsorbents were able to sustain substantial uranium adsorption over several cycles. The structure property correlations of polymeric amidoxime adsorbents were further examined which helped in discovering that the accessibility of binding sites and the density of amidoxime groups both influenced performances. While well distributed ligands maximized U(VI) adsorption, over functionalization may decrease adsorption due to steric hindrance. Additionally, the study verified that amidoxime adsorbents have great selectivity and quick adsorption kinetics, which makes them ideal for real world uses [65]. Amidoxime functionalized fibers were investigated for their advantages in mass transfer and quick kinetics. Increased surface area and better amidoxime group accessibility for uranyl ions were offered by the fibrous shape. The resilience of amidoxime functionalization for real world uranium recovery applications was confirmed by adsorption tests that showed great selectivity and efficiency even in the presence of competing ions [66].

4.1.3 Mechanistic insights

The main cause for best adsorption performance of amidoxime functionalized materials comes mainly from bidentate chelation between the uranyl ion (UO_2^{2+}) and the amidoxime group, where the oxime oxygen and imine nitrogen coordinate to the metal center to form stable five membered rings as shown in the Fig. 3 [51]. Because U(VI) is a hard Lewis acid that preferentially binds to hard Lewis's base donor atoms like oxygen and nitrogen, this inner sphere complexation is greatly favoured [52]. Because GO has a large surface area and many anchoring sites, amidoxime ligands are more accessible and

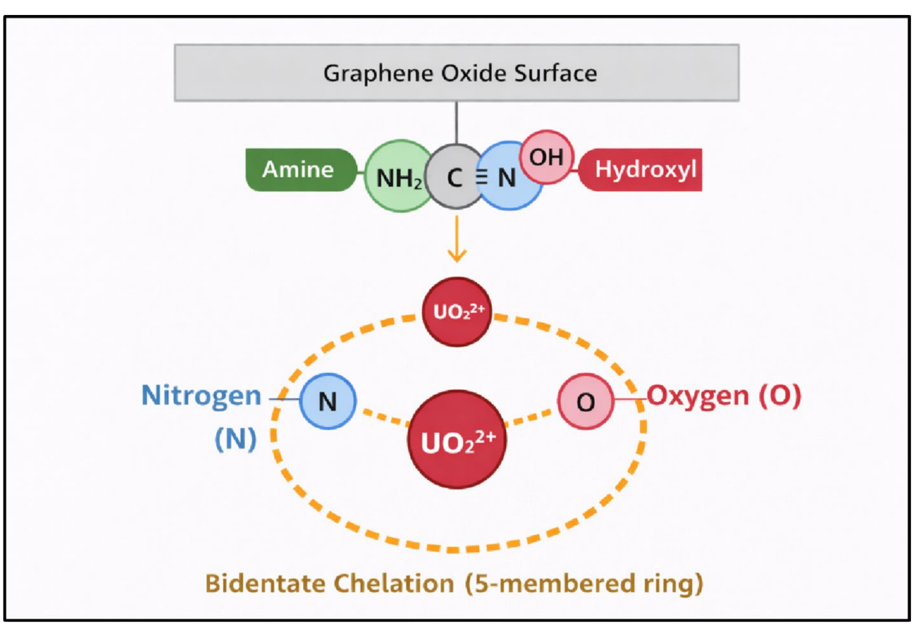


Fig. 3 Binding mechanism of amidoxime functionalization

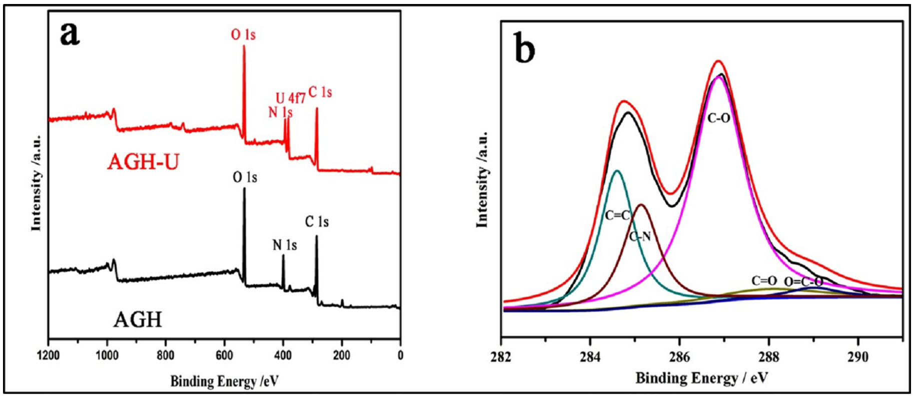


Fig. 4 XPS spectra of **a** AGH and **b** AGH after uranium adsorption. Reproduced with the permission from [64]. Available under a CC-BY 4.0 license. Copyright 2016 nature.

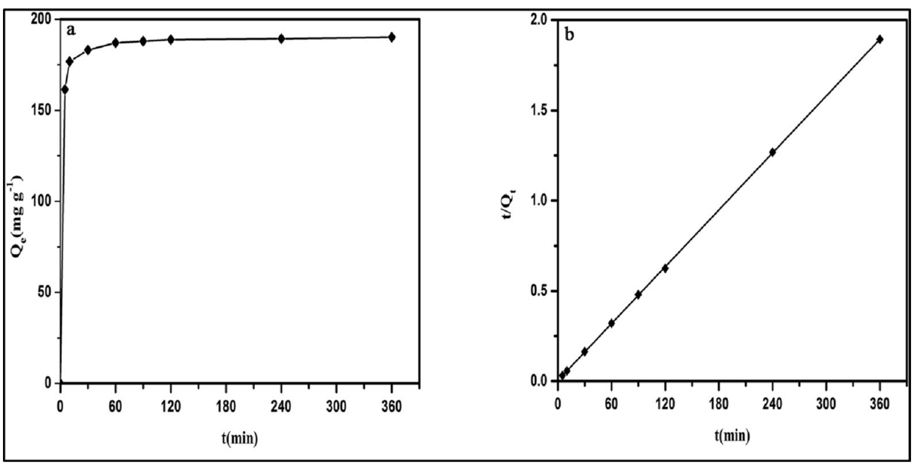


Fig. 5 **a** Effect of contact time on uranium (VI) adsorption. **b** pseudo-second-order model for the adsorption of uranium (VI) on AGH. Reproduced with the permission from [64]. Available under a CC-BY 4.0 license. Copyright 2016 Nature

widely distributed, improving uranyl binding efficiency [55]. Preventing sheet restacking improves mass transfer and adsorption kinetics in 3D graphene-based hydrogel frameworks by increasing the accessibility of amidoxime sites [62]. Because uneven or excessively dense functionalization can result in steric hindrance, which lowers the availability of active sites for uranyl ions, the efficiency of adsorption also depends on the distribution of ligands [63]. Research on GO/amidoxime hydrogels verified that U(VI) forms inner sphere complexes by directly binding to the nitrogen and oxygen atoms of the amidoxime groups which can be identified by Fig. 4. It shows the XPS data of AGH before and after uranium adsorption, while the pseudo-second-order model fit Fig. 5(a). confirms that the process is controlled by stable chemical chelation between uranyl ions and the amidoxime functional groups, and the adsorption kinetics Fig. 5(b). show rapid capture, with over 90% of uranium removed in 60 min [64]. Even in the presence of competing ions including Na^+, Mg^{2+}, and Ca^{2+}, polymeric amidoxime adsorbents maximize adsorption capacity and selectivity by high ligand density and suitable spatial distribution [65]. Fiber based amidoxime adsorbents retain the bidentate chelation process as the predominant mode of U(VI) binding while achieving fast adsorption kinetics through improved mass transfer [66].

4.1.4 Limitations of amidoxime functionalization

Amidoxime functionalized adsorbents have a number of drawbacks that limit their practical use, despite their great selectivity and adsorption capacity. The availability of donor sites for U(VI) binding can be decreased by protonation of the amidoxime nitrogen or oxime oxygen in acidic settings, and adsorption efficiency can also be impacted by extremely alkaline conditions [51]. This is one significant constraint. A key factor is the structural arrangement of functional groups. Steric hindrance, which prevents uranyl from accessing binding sites and lowers overall adsorption effectiveness, might result from excessive grafting or an uneven distribution of amidoxime ligands [52, 55]. Long term durability in graphene-based hydrogels may be limited by partial aggregation or structural collapse during repeated adsorption-desorption cycles, even if 3D designs enhance site accessibility [62].

Additionally, amidoxime groups are susceptible to slow hydrolysis and oxidation when exposed to water for extended periods of time, which can progressively lower the number of active sites and effectiveness over several cycles [63]. Competition from numerous cations in seawater, such as Na^{+}, Mg^{2+}, and Ca^{2+}, might disrupt uranyl binding for polymeric adsorbents, necessitating high ligand density and optimal functionalization for practical recovery [64, 65]. Lastly, acidic or harsh desorption conditions are frequently used in the regeneration of amidoxime adsorbents, which may partially deteriorate the material and risk its mechanical stability [66].

4.2 Phosphoryl/phosphonate functionalization

Phosphoryl (PO_3H_2) and phosphonate (PO_3^{2-} or PO_3R^{2-}) functional groups have been extensively studied as highly selective ligands for uranium removal from aqueous environments due to their strong affinity toward uranyl ions. Their efficiency comes from the presence of many oxygen donor atoms that can form stable inner sphere coordination complexes with UO_2^{2+}. The uniform grafting of these ligands on GO results in great accessibility of active sites [68]. And the addition of phosphonate groups with flexible $-CH_2-$ spacers improve ligand density and coordination efficiency, resulting in larger adsorption capacities and faster kinetics [69]. The higher electron density of phosphoryl oxygen atoms improves uranyl binding and reduces interference from competing ions, as shown by Zhang et al. [70] using phosphoryl functionalized GO. Also, the carbon framework provides structural stability, prevents ligand aggregation, and allows uniform ligand dispersion, which enhances reusability and adsorption efficiency across repeated cycles, as demonstrated in phosphate modified carbon materials by Hu et al. [71]. Through multisite oxygen donation and synergistic interactions with large surface area carbon supports, phosphoryl and phosphonate ligands to achieve strong, selective, and stable uranyl coordination under environmentally relevant situations as shown by all of these studies. This results in high adsorption capacities, fast kinetics, and excellent selectivity.

4.2.1 Mechanistic insights

The adsorption of uranium onto phosphoryl or phosphonate functionalized materials is mainly controlled by chemical coordination between uranyl ions and the oxygen

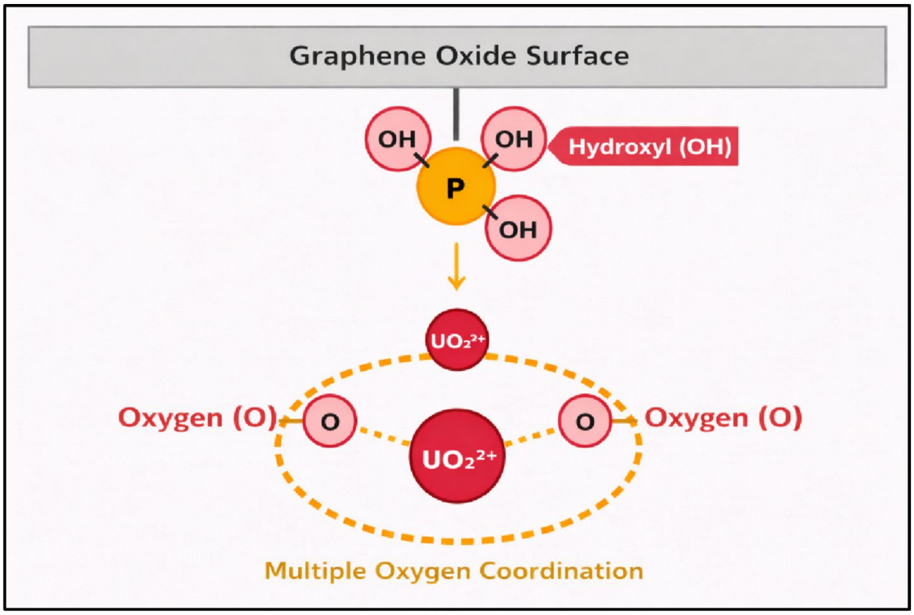


Fig. 6 Binding mechanism of phosphoryl/phosphonate functionalization

donor atoms of the functional groups. Fig. 6. demonstrates the binding mechanism of phosphoryl functionalization [68]. From the experiment of Yuan et al. [69], the uranyl ion's O = U=O structure permits direct interaction with P = O and P–OH molecules to create inner sphere surface complexes, a binding mechanism with significant covalent character that leads to strong and selective adsorption. As demonstrated in phosphoryl functionalized GO by Zhang et al. [70], the high surface area carbon and GO supports offer a stable framework that ensures uniform dispersion of functional groups and inhibits ligand aggregation, thereby maintaining the accessibility of coordination sites. The planar two-dimensional structure of GO further enables multisite coordination, where a single uranyl ion simultaneously interacts with multiple phosphoryl or phosphonate groups, improving both binding strength and selectivity, as demonstrated by phosphonate modified GO by Yuan et al. [69]. However, Zhang et al. [70] specifically mentioned that excessive ligand loading may result in steric crowding and lower adsorption effectiveness. Adsorption is thermodynamically favourable at moderately acidic to near neutral pH, where reduced phosphate or phosphonate groups easily coordinate with U(VI). As demonstrated and verified for phosphoryl functionalized GO equilibrium is reached kinetically in two to three hours due to strong chemisorption at ligand sites and fast diffusion on the carbon framework.

Phosphoryl and phosphonate functionalized carbon-based adsorbents, especially GO have become as a very efficient materials for uranium recovery due to their strong attraction toward uranyl ions (UO_2^{2+}) and coordination with oxygen donor atoms in P = O and P-OH groups which can be visually seen in the results of SEM analysis and FTIR spectra from the Figs. 7 and 8 and a new P 2p peak at 137.4 eV and a C-P peak at 285.2 eV, indicating that the phosphorus was successfully grafted onto the carbon framework. A P-O stretching vibration at 1640 cm^{-1} and distinctive phosphate bands at 600, 960, and 1020 cm^{-1} were identified using complementary Fourier transform infrared (FT-IR) spectroscopy [68]. High surface area, structural stability, and even dispersion of active sites is all provided by the carbon framework. Phosphate functionalized GO yielded an adsorption

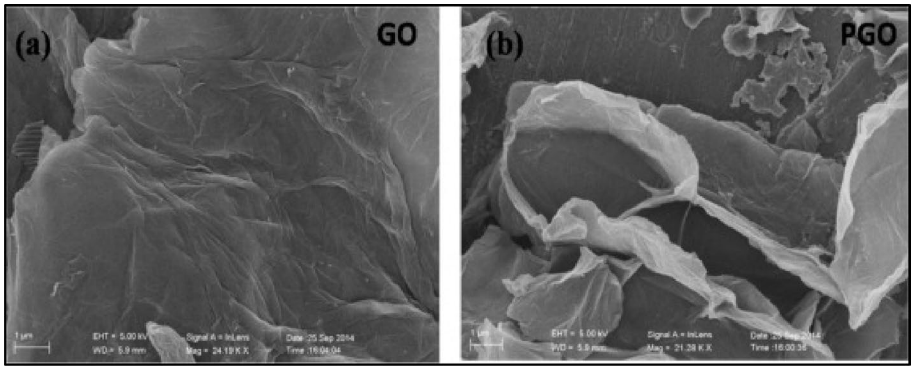


Fig. 7 SEM images of GO (**a**) and PGO (**b**) [68]. Reproduced with the permission from [68]. Copyright 2015 Elsevier.

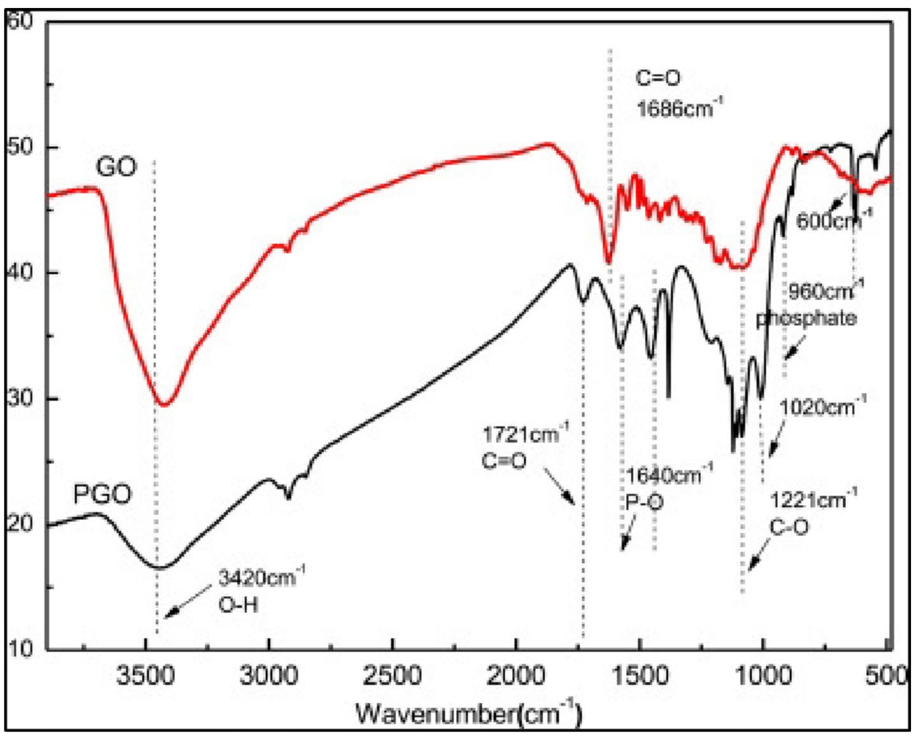


Fig. 8 The FT-IR spectra of PGO and GO. Reproduced with the permission from [68]. Copyright 2015 Elsevier

capacity of about 251.7 mg·g^{-1} at pH 5–6 with fast adsorption kinetics that followed a pseudo second order model, suggesting that chemisorption controls the adsorption process. The adsorbent was highly selective for uranyl ions despite competing cations such Na^+, Ca^{2+}, and Mg^{2+}. However, adsorption effectiveness decreased at very acidic conditions due to protonation of phosphate groups and lower ligand availability [69]. In order to improve ligand accessibility and reduce steric effects, Yuan et al. synthesized phosphonate modified GO with flexible $–CH_2–$ spacers. This led to a greater adsorption capacity of 320 mg·g^{-1} and equilibrium attainment in about 120 min. The material maintained more than 85% of its original capacity after several adsorption–desorption cycles, indicating outstanding chemical stability and reusability, and its adsorption data fit the Langmuir isotherm well, confirming monolayer adsorption. Phosphoryl functionalized GO with a capacity of 290 mg·g^{-1} was reported by Zhang et al. [70] They attributed the improved performance to the high electron density of phosphoryl oxygen atoms, which enable robust uranyl coordination. Although high functional group loading resulted in mild steric hindrance that prevented further increases in adsorption capacity, this system showed little interference from competing ions. Depending on surface area and functional group density, Hu et al.'s investigation of phosphate modified carbon materials yielded adsorption capabilities ranging from 200 to 270 mg·g^{-1}. While adsorption efficiency decreased in the presence of strongly competing multivalent ions or under extreme pH conditions, their analysis demonstrated good selectivity and steady performance over multiple cycles [71].

The mechanism of uranyl uptake involves the formation of stable covalent like bonds that promote high thermodynamic stability through inner sphere complexation between UO_2^{2+} and phosphoryl or phosphonate oxygen atoms [68]. By limiting aggregation, facilitating multisite coordination, and enhancing adsorption kinetics, the carbon support further improves performance [69]. All things considered, phosphoryl and phosphonate functionalized adsorbents are very promising materials for uranium recovery from contaminated water and nuclear waste streams because they combine high adsorption capacity, fast kinetics, excellent selectivity, and good reusability [70]. However, further optimization is needed to address pH sensitivity, steric limitations, and synthesis complexity for large scale deployment [71].

4.2.2 Advantages of phosphoryl/phosphonate functionalization

Phosphoryl and phosphonate functionalized carbon-based adsorbents offer considerable benefits for uranium recovery, principally due to their strong and selective affinity toward uranyl ions (UO_2^{2+}) from oxygen rich donor environments. P = O and P–OH groups allow uranyl ions to form stable inner sphere complexes that have strong binding energies and large adsorption capacities. The chemical sensitivity of phosphate groups toward U(VI) was highlighted by the quick adsorption kinetics and strong selectivity even in the presence of competing alkali and alkaline earth metal ions [68]. According to Yuan et al., phosphonate functionalized GO further enhances adsorption performance by introducing flexible alkyl linkers that improve ligand accessibility and remove steric limitations, resulting in larger adsorption capacities and faster equilibrium periods. Phosphoryl based adsorbents' excellent structural and chemical stability, which enables them to maintain a sizable portion of their adsorption capacity throughout several regeneration cycles, is another significant benefit that makes them appropriate for repeated usage in real world applications [69]. According to Zhang et al., phosphoryl functionalized GO exhibits resilience in challenging aqueous environments by maintaining good selectivity and adsorption effectiveness over a wide range of ionic strengths [70]. Also, as demonstrated by Hu et al., phosphate modified carbon materials benefit from synergistic effects between surface area and functional group density, permitting effective uranium uptake while preserving good mechanical integrity and reusability. High adsorption capacity, quick kinetics, strong selectivity, outstanding reusability, and structural robustness are the main benefits of phosphoryl and phosphonate functionalization, which make these materials very promising for uranium extraction from contaminated water and nuclear waste streams [71].

4.2.3 Limitations and challenges of phosphoryl/phosphonate functionalization

Phosphoryl and phosphonate functionalized carbon-based adsorbents have a strong affinity for uranyl ions, but they have a number of intrinsic limitations that may limit their performance in real world situations. One significant limitation is their sensitivity to pH, as adsorption efficiency drastically decreases under strongly acidic conditions due to protonation of P–OH groups, which reduces the availability of oxygen donor sites for uranyl coordination, as seen for phosphate functionalized GO [68]. In highly alkaline environments, competitive complexation with carbonate species can also suppress uranium uptake, which limits their efficiency in natural or seawater systems [69]. As observed for phosphoryl functionalized GO [70], steric hindrance resulting from excessive functional group loading presents another difficulty. High densities of phosphoryl groups may prevent access to adsorption sites and reduce effective surface area, resulting in decreased adsorption capacity. Furthermore, because of their great affinity for phosphate ligands, highly competing metal ions like Fe^{3+} or Al^{3+} can partially interfere with uranyl binding, decreasing selectivity in complex aqueous matrices [71].

Although generally good, reusability is not completely loss free because performance might deteriorate over time due to phosphoryl or phosphonate groups gradually degrading or leaching over repeated adsorption–desorption cycles [69]. Also, multistep chemical procedures are frequently used in the synthesis of consistently functionalized, well dispersed phosphoryl modified carbon materials, which might raise costs and restrict large scale manufacturing [70]. All of these drawbacks show that in order to effectively utilize the potential of phosphoryl and phosphonate functionalized adsorbents for uranium recovery, additional tuning of functional group density, structural stability, and operating conditions is required [71].

4.3 Amino functionalization

Amino functionalization increases the efficiency of carbon-based materials for uranium extraction by combining nitrogen donor atoms with uranyl ions (UO_2^{2+}). Amino groups like primary amines (–NH_2), secondary amines (–NH–), and tertiary amines (–N–) can be added to GO, rGO, activated carbon, and other carbon nanomaterials to produce active sites that can bind with uranium ions in aqueous environments. Liu et al. claim that amino functionalized GO which had a greater uranium adsorption capacity than pure GO, was produced by amide interactions between GO carboxyl groups and amine containing molecules [72]. By adding pyrrolic and pyridinic nitrogen functionalities that functioned as coordination centers, Jin et al. demonstrated that nitrogen doped GO produced by hydrothermally processing nitrogen rich precursors had a significantly greater affinity for U(VI) [73]. It is also shown that the higher density of electron donating nitrogen sites in nitrogen doped carbon materials produced by thermal treatment in ammonia atmospheres resulted in better adsorption kinetics and more selectivity toward U(VI). Together, these experiments demonstrated that amino and nitrogen functionalities can be logically added to carbon frameworks to improve uranium binding performance without sacrificing the adsorbent's inherent surface area or mechanical stability [74]. In order to increase selectivity and stability under ecologically relevant conditions, more recent research has concentrated on ligand assisted amino functionalization. Liu et al. created cysteine modified GO by adding both amino and thiol groups, which improved resistance to interference from coexisting metal ions and increased uranium adsorption capacity [75]. Thiol functionalized GO with amino moieties was created by Liu et al. and shown synergistic coordination effects that enhanced uranium adsorption and recyclability. These results demonstrate the increasing interest in amino based surface engineering as a viable method for creating uranium adsorbents with good performance [76].

4.3.1 Mechanistic insights

Uranyl ion adsorption on amino functionalized carbon materials is principally driven by coordination interactions between nitrogen donor atoms and the uranium center, which are enhanced by electrostatic attraction and surface complexation has been demonstrated in the

Figure 9. Amino groups can form stable inner sphere complexes by donating electron density to the vacant orbitals of U(VI) through their lone pair electrons on nitrogen. Strong pH dependent adsorption behavior was shown by Liu et al. for amino functionalized GO, with maximum uranium uptake taking place at near neutral pH, when amino groups are largely reduced and available for coordination [72]. According to Jin et al., strong covalent like interactions that control the adsorption mechanism are made possible by nitrogen doped GO interacting with uranyl ions through pyridinic nitrogen sites that act as Lewis's bases [73]. Also, nitrogen doped carbons have increased electron density on surface nitrogen atoms, which enhances thermodynamic favourability and promotes stronger metal-ligand coordination [74]. Amino groups frequently work in concert with other functional moieties in multifunctional systems to improve adsorption performance. According to Liu et al., cysteine modified GO increases adsorption capacity and improves resistance to competitive ions by forming chelate like complexes with uranyl ions through cooperative binding involving amino and thiol groups [75]. Thiol functionalized GO with amino groups showed multisite coordination behavior, which encouraged better binding and slower uranium species desorption [76].

Multiple studies have consistently demonstrated that adding amino functionalities to carbon-based adsorbents improves uranium adsorption capacity, kinetics, and

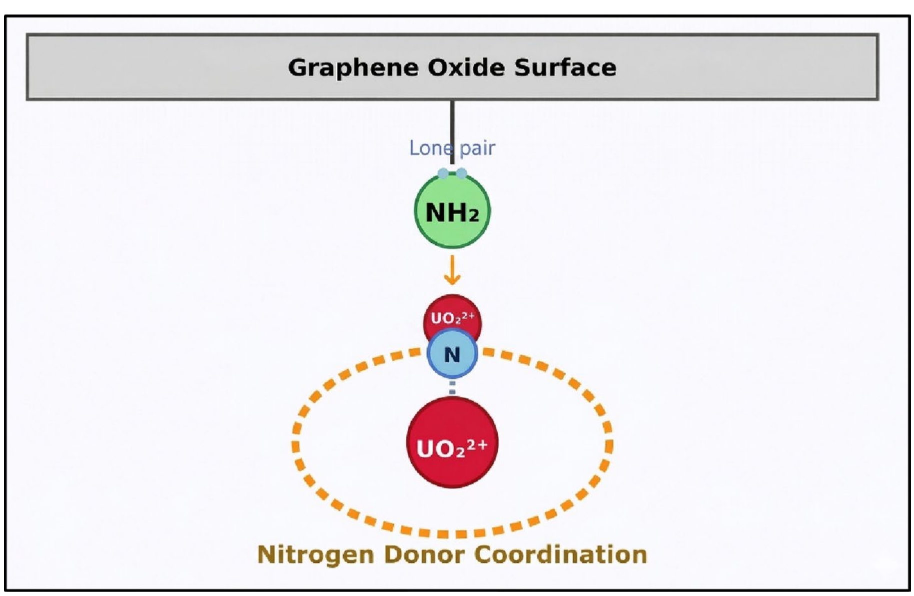


Fig. 9 Binding mechanism of amino functionalization

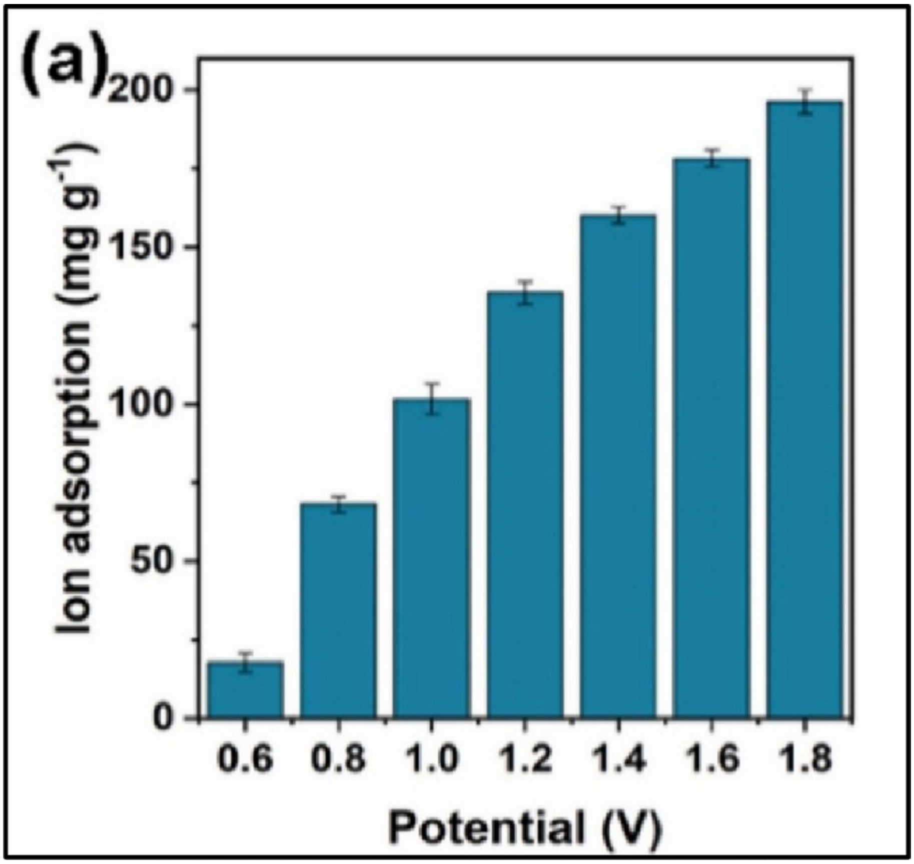


Fig. 10 **a** Ion adsorption capacity [73]. Reproduced with the permission from [73]. Copyright 2023 Elsevier

selectivity. According to Liu et al., at pH 5–6, amino functionalized GO had an adsorption capacity of roughly 180 mg·g^{-1}, which was much greater than that of unmodified GO. The uranium uptake mechanism was dominated by chemisorption, as shown by the pseudo second order kinetic model of the adsorption process [72]. Jin et al. showed that nitrogen doped GO quickly reached equilibrium in 90 min and had a uranium adsorption capacity of about 194 mg·g^{-1} demonstrated in Fig. 10. A type IV N_2 adsorption-desorption isotherm with a distinct hysteresis loop can be seen in Fig. 11., indicating the presence of tiny mesopores, characterizes the unique pore structure of Mn and N co-doped carbon nanospheres (Mn-NC). Because of its enriched 3D electronic structure, Mn-NC performs better in capacitive deionization (CDI) than metal-free nitrogen-doped carbon (NC) despite having a much lower specific surface area of 20.7 m^2/g and no micropores. The Langmuir isotherm model provided a good fit to the adsorption data, indicating monolayer adsorption on uniformly distributed active sites [73]. Nitrogen doped carbon materials showed capacities more than 230 mg·g^{-1}, which were explained by higher nitrogen content and improved electron donating capacity [74]. According to Liu et al., coordinated binding actions combining amino and thiol groups allowed cysteine modified GO to attain an adsorption capacity of about 260 mg·g^{-1} [75]. Thiol functionalized GO with amino moieties reached capacities of about 240 mg·g^{-1} and had good recyclability across several cycles of adsorption and desorption [34].

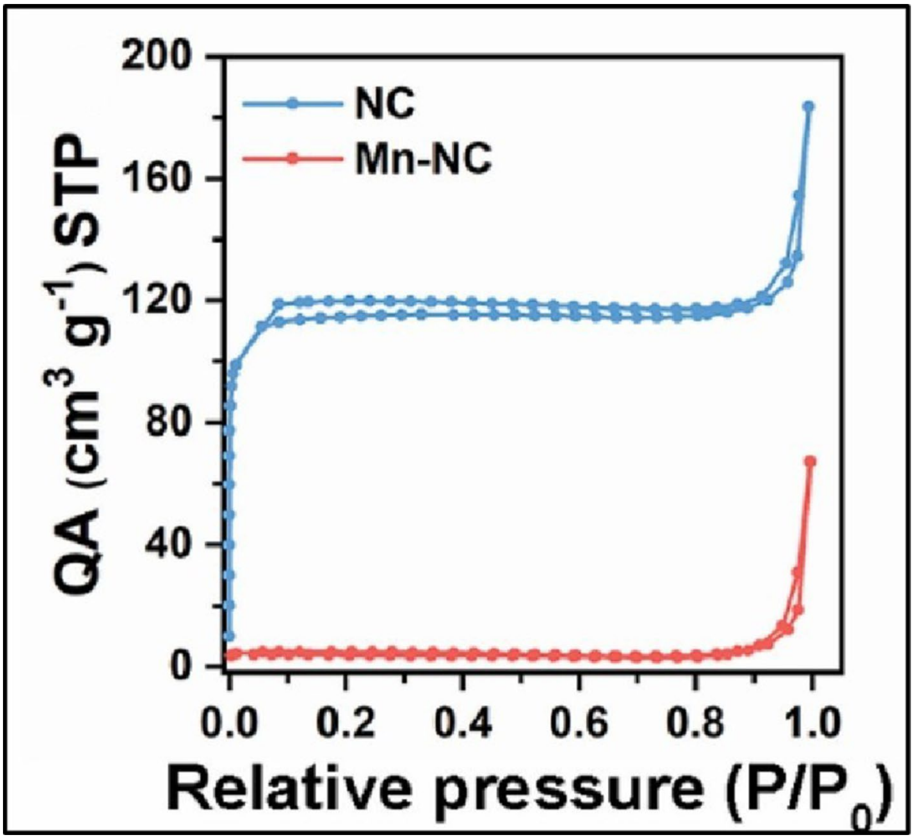


Fig. 11 N_2 adsorption-desorption curves of NC and Mn-NC samples [73]. Reproduced with the permission from [73]. Copyright 2023 Elsevier

4.3.2 Advantages of amino functionalization

For uranium extraction applications, amino functionalization provides several significant benefits. Amino groups offer strong coordination sites that form stable inner sphere complexes with uranyl ions, leading to enhanced adsorption capacity [72]. The increased surface reactivity and electron donating nature of nitrogen doped and amino functionalized materials result in faster adsorption kinetics [73]. Owing to favourable coordination geometry and electronic compatibility, amino functionalized adsorbents exhibit high selectivity for U(VI) over competing metal ions [74]. Cooperative binding effects in multifunctional amino based adsorbents further improve adsorption stability and overall performance [75]. Amino functionalized materials also retain their surface area and structural integrity over multiple adsorption-desorption cycles, reflecting the strong covalent attachment of nitrogen containing ligands [76].

4.3.3 Limitations and challenges of amino functionalization

Amino functionalized adsorbents have several drawbacks despite their excellent performance. pH sensitivity is a significant problem since amino groups become less available for uranyl coordination when they are protonated in acidic environments [72]. Reduced adsorption effectiveness in highly alkaline conditions, where uranyl speciation and surface charge effects restrict binding interactions, is another drawback [73]. At high amino ligand densities, steric hindrance can restrict maximal adsorption capacity and decrease accessible surface area [74]. Uranium uptake in complicated aqueous environments may be limited by competitive adsorption from nearby metal ions, especially those with a high affinity for nitrogen donors [75]. Because repeated adsorption-desorption cycles might cause amino ligands to gradually degrade or leach, long term stability is still an issue [76].

4.4 Dual functionalization

Dual functionalization is an advanced surface engineering strategy for uranium adsorbents that integrates two complementary ligands onto a single carbon framework to improve adsorption capacity, selectivity, and stability. Dual functionalized materials, as opposed to single ligand systems, offer uranyl ions multiple coordination environments, enabling cooperative binding and multisite complexation that greatly enhances adsorption performance under challenging aqueous conditions. The advantage of combining nitrogen rich ligands with polymeric supports was demonstrated by the synthesis of amidoxime-chitosan GO hydrogels with both amidoxime and amino functionalities and reported improved uranium adsorption behavior compared with mono functional analogues [77]. By grafting both chelating amidoxime groups and highly coordinating phosphoryl groups onto the GO surface, Li et al. created amidoxime-phosphoryl functionalized GO, a multifunctional ligand environment that can bind uranyl ions with remarkably high affinity [78]. By coupling imine groups with amidoxime ligands, Schiff base reinforced amidoxime GO, which enhanced adsorption endurance under repeated cycling conditions and further stabilized the ligand architecture [79]. By combining two chelating ligands that may form bidentate and multidentate complexes with U(VI), poly (amidoxime-hydroxamic acid) modified GO was developed, which greatly increased adsorption capacity and selectivity [80]. The coexistence of numerous heteroatoms improves uranium binding and structural endurance, as demonstrated by the preparation of oxygen and nitrogen rich GO aerogels with both amino and oxygenated groups [86]. By combining two chelating ligands that may form bidentate and multidentate complexes with U(VI), poly (amidoxime-hydroxamic acid) modified GO was developed, which greatly increased adsorption capacity and selectivity. When taken as a whole, these results show that dual functionalization is an effective method for creating next generation uranium adsorbents with better performance and operational stability [87].

4.4.1 Mechanistic insights

The increased performance of dual functionalized adsorbents is due to cooperative coordination mechanisms in which uranyl ions interact with two different ligand types at the same time, resulting in stable chelate-like structures and multisite inner sphere complexes. In contrast to single amidoxime systems, amidoxime chitosan GO hydrogels shown binding towards U(VI) by cooperative N-O interaction involving both amidoxime nitrogen atoms and nearby hydroxyl groups. Figure 12. demonstrates the binding mechanism of Amino + carboxyl dual functionalization. Amidoxime phosphoryl GO forms highly stable bidentate complexes with uranyl ions, where phosphoryl oxygen atoms provide strong electrostatic and covalent interactions while amidoxime groups stabilize the coordination geometry, producing high thermodynamic stability [77]. Schiff base Reinforced amidoxime GO enhances ligand rigidity and promotes multidentate binding, which reduces ligand dissociation and improves resistance to competitive ions [78]. Figure 13. demonstrates the binding mechanism of Amidoxime + Phosphoryl

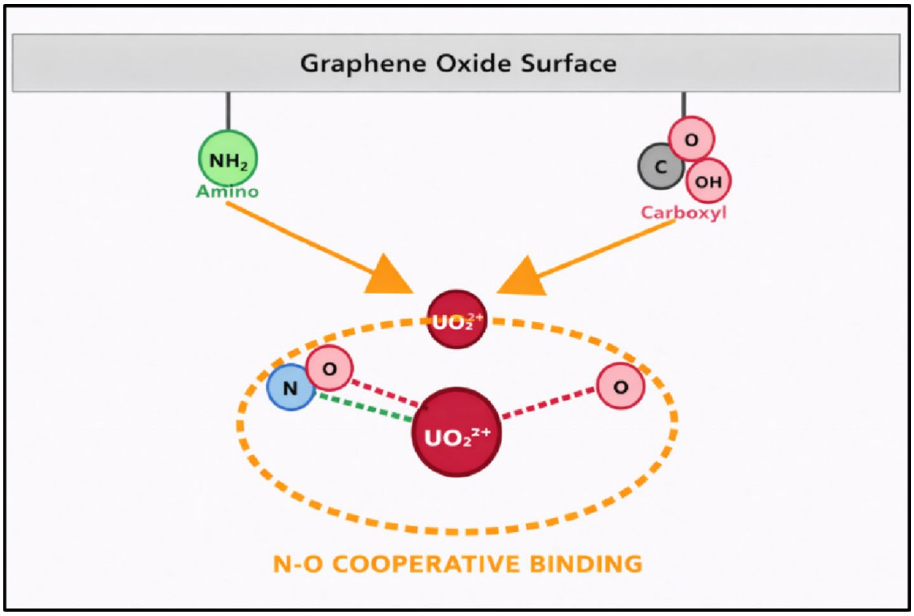


Fig. 12 Binding mechanism of dual functionalization (amino + carboxyl)

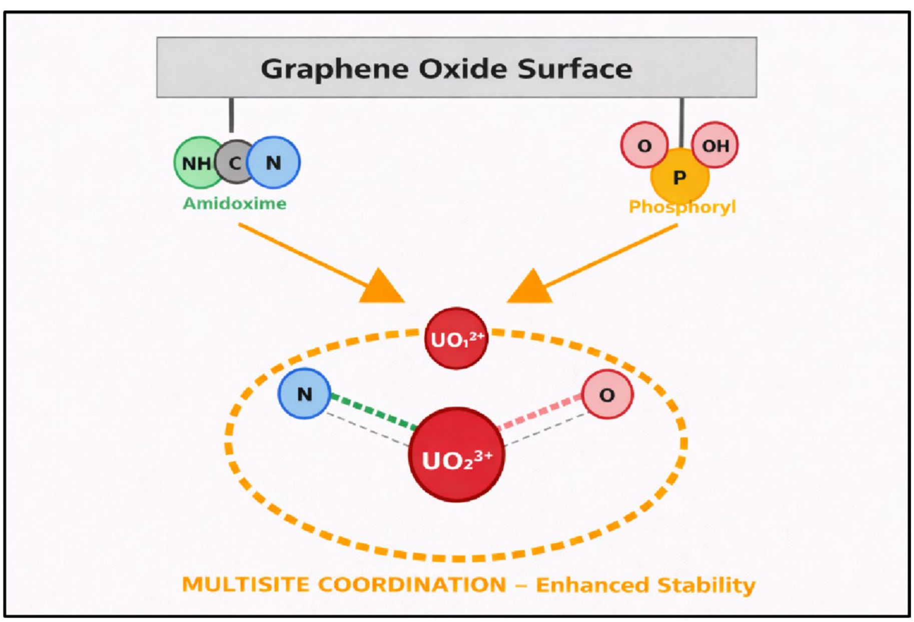


Fig. 13 Binding mechanism of dual functionalization (amidoxime + phosphoryl)

dual functionalization. According to Jabbar et al., oxygen nitrogen rich GO aerogels improve binding strength and adsorption kinetics by facilitating multisite coordination through simultaneous interactions [79]. According to Zhu et al. [80], poly (amidoxime–hydroxamic acid) GO creates extremely strong surface complexes that are resistant to desorption by forming stable chelation rings with U(VI) by cooperative N-O coordination [86]. Amino thiol oxygen GO materials in more advanced systems, where uranyl ions are stabilized by three cooperative donor atoms, resulting in remarkable kinetic performance and adsorption affinity [87].

According to experimental findings, dual functionalized adsorbents perform better than mono functional systems in terms of stability, selectivity, and adsorption capacity. Amidoxime chitosan GO hydrogels achieved uranium adsorption capabilities of about 280 mg·g^{-1}, which were much higher than those of pure GO and single amidoxime functionalized materials. Chemisorption dominated uptake was confirmed by the pseudo second order model of the adsorption kinetics. Amidoxime phosphoryl GO to achieve an extraordinarily high adsorption capacity of about 350 mg·g^{-1}, which was explained by the high surface area of the GO support and the two ligands' synergistic chelation effect [77]. The Langmuir, Sips, and Freundlich isotherm models which together explain the link between the amount of uranium adsorbed and its equilibrium concentration fit the data which is demonstrated in the Fig. 14. Specifically, the Freundlich and Sips fits show that multilayer sorption and surface heterogeneity also play roles, but the fit to the Langmuir model supports monolayer sorption, in which uranium forms a single layer on particular active sites. Additionally, the kinetics exhibit a pseudo-second-order model, indicating that chemisorption which involves strong chemical interactions between the uranium ions and the functional amidoxime and phosphate groups is the limiting phase. Schiff base reinforced amidoxime GO showed exceptional structural and chemical stability with an adsorption capacity of over 300 mg·g^{-1} and kept over 90% of its capacity after five adsorption desorption cycles [78]. Because of the porous aerogel structure, Jabbar et al. [79] demonstrated that oxygen nitrogen rich GO aerogels demonstrated fast adsorption kinetics and reached capacities of about 260 mg·g^{-1}. Poly (amidoxime-hydroxamic acid) GO demonstrated excellent selectivity in the presence of competing ions including Ca^{2+} and Mg^{2+}, achieving adsorption capacities close to 340 mg·g^{-1} [86].

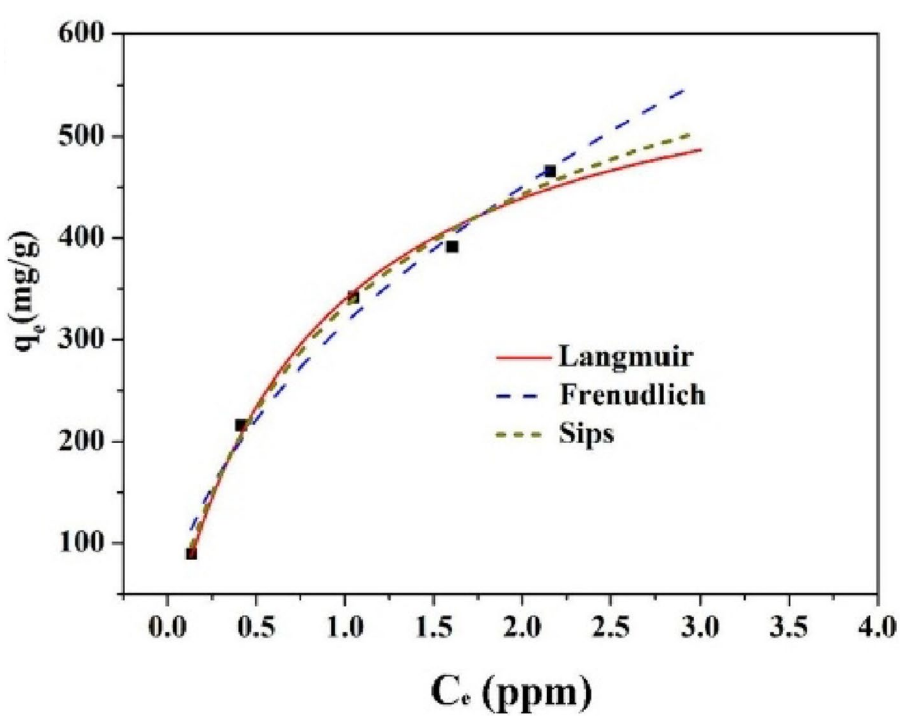


Fig. 14 Fitted with Langmuir, Freundlich and Sips isotherm models [78]. Reproduced with the permission from [78] Copyright 2024 Elsevier

Amino thiol oxygen GO materials have capacities of about 320 $mg \cdot g^{-1}$ and to be highly recyclable over several cycles [86].

4.4.2 Advantages of dual functionalization

Dual functionalization offers various advantages for uranium extraction applications [77]. Thermodynamic stability and binding strength are significantly increased when two ligand types work together [78]. By providing several active centers for each uranyl ion, multisite binding increases adsorption capacity [79]. By creating ligand environments specific to uranyl coordination geometry, dual functional materials exhibit improved selectivity [80]. Synergistic ligands sustain high performance and boost resistance to competing ions in complex aquatic systems [86]. Dual functional designs decrease ligand degradation and increase recyclability during multiple adsorption desorption cycles [87].

4.4.3 Limitations and challenges of dual functionalization

Dual functionalized adsorbents have a number of limitations despite their better performance [77]. Scalability is limited and production costs are increased due to the complex and multi-step synthesis procedure [78]. It is still challenging to precisely manage the ligand ratio and spatial distribution, which might result in uneven active sites and decreased repeatability [79]. Maximum adsorption capacity may be limited and available binding sites may be reduced due to steric hindrance between closely spaced ligands [80]. In actual water matrices, active centers may nevertheless be partially blocked by competitive adsorption and fouling [86]. Insufficient research has been done on long term stability in extremely acidic, alkaline, or radiologically hostile environments [87].

4.5 Tri functionalization

Tri functionalization is an advanced ligand engineering method that involves the intentional incorporation of three distinct chemically different functional groups onto a single adsorbent material to improve adsorption capacity, selectivity, and stability. Unlike single or dual functionalized systems, tri functional systems are designed to imitate the multidentate coordination spheres characteristic of natural, more effective chelating ligands, enhancing the binding interaction with uranyl ions. An amino acid assisted multifunctional GO combining amine, carboxyl, and hydroxyl functionalities, revealing that cooperative interactions among these groups considerably improve uranyl adsorption and structural resilience [88]. By creating a tri functional ligand modified GO with binding sites based on nitrogen, oxygen, and phosphorus helped in achieving remarkable uranium selectivity even in extremely competitive ionic conditions [89].

These studies emphasize tri functionalization as a logical design strategy to get around the inherent drawbacks of more straightforward functionalization techniques [88]. GO improves performance in complex aqueous systems related to environmental remediation and nuclear wastewater treatment by introducing three complementing functional groups that produce a heterogeneous yet coordinated binding environment [89].

4.5.1 Mechanistic insights

Cooperative inner sphere complexation between uranyl ions and many donor atoms dispersed throughout the GO surface dominates the adsorption mechanism in tri

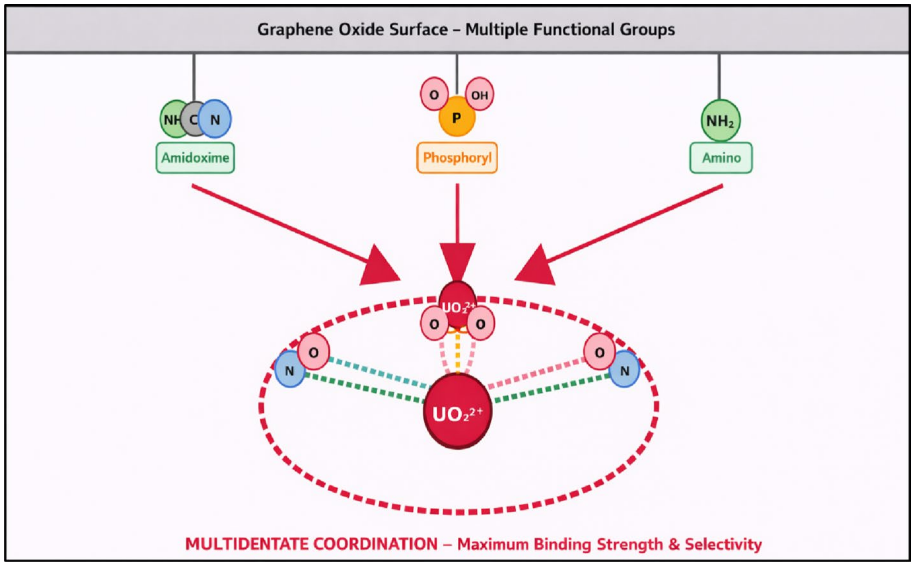


Fig. 15 Binding mechanism of tri functionalization (amidoxime + phosphoryl + amino)

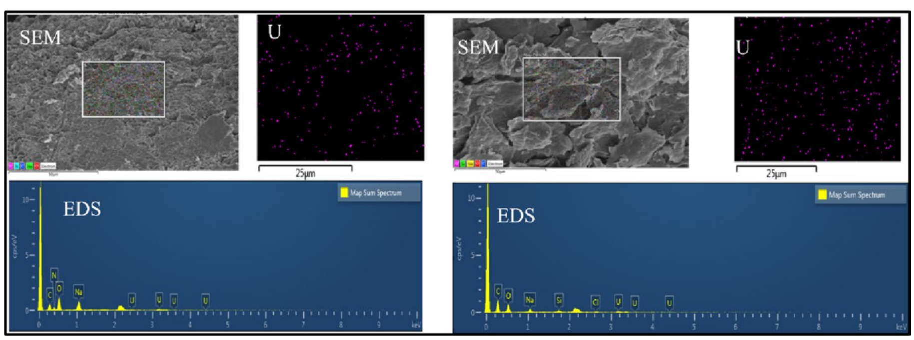


Fig. 16 SEM-EDS mapping of U(VI)-loaded GO-CS-AO (left), GO-CMC-AO (right) of selected area [88]. Reproduced with the permission from [88]. Copyright 2024 Springer Nature

functionalized GO systems. In amino acid assisted multifunctional GO, hydroxyl groups provide secondary hydrogen bond stabilization, while uranyl binding takes place by simultaneous coordination with amino nitrogen atoms and carboxylate oxygen atoms to produce a very stable chelation structure. More binding strength is produced by this multisite interaction than by separate functional groups [88]. Figure 15. explains the binding mechanism of a tri functionalization where amidoxime, phosphoryl and Amino groups are integrated. Phosphoryl or phosphonate oxygen atoms serve as the main hard base donors to UO_2^{2+} in tri functional ligand engineered GO, whereas amine groups improve electrostatic attraction and enable uranyl ion preconcentration close to the surface. Through further coordination or hydrogen bonding, carboxyl or hydroxyl groups further stabilize the adsorbed molecule, producing a dense and highly structured adsorption layer [89]. One uranyl ion can interact with two or more ligands at once thanks to GO's planar structure, which encourages spatial closeness among functional groups and greatly improves thermodynamic stability. The production of inner sphere complexes with partial covalent character, which explains the remarkable selectivity and resistance to competitive ion interference observed experimentally, is confirmed by density functional theory. Scanning Electron Microscopy (SEM) photos show the microscopic morphology of the gels, whereas Energy Dispersive Spectroscopy (EDS) spectra shown in Fig. 16. and elemental mapping provide visual and chemical confirmation of the elements present. In particular, the mapping images show that uranium ions are evenly dispersed across both gels' structures. This homogeneous distribution, together with the primary oxygen and carbon components found in the spectra, demonstrates that the functionalized active sites successfully absorbed the U(VI) ions along the adsorbents' whole surface [88, 89].

Both studies experimental findings show that, in comparison to mono or dual functional GO, tri functionalization significantly improves uranium adsorption performance. Chen et al. showed that amino acid assisted multifunctional GO obtained significantly greater adsorption capacities and faster kinetics due to synergistic ligand interactions and increased dispersion of functional groups over the GO surface. Multiple functional moieties kept active sites accessible and avoided ligand overpopulation, which is frequently a drawback in highly functionalized single ligand systems. The significance of multisite coordination in preventing nonspecific adsorption has been demonstrated by tri functional ligand engineered GO demonstrated remarkable selectivity for U(VI) even in the presence of excess competing ions including Ca^{2+}, Mg^{2+}, and Na^{2+}. The adsorption procedure continued to be successful over a comparatively broad pH range, suggesting that tri functional systems have improved chemical flexibility. Also, after several adsorption-desorption cycles, both tests showed good regeneration performance with no loss of adsorption capacity remaining at a level lower than 10% after four consecutive cycles as shown in Fig. 17., which is indicative of the structural stability provided by the multifunctional ligand design. This consistent performance demonstrates that the

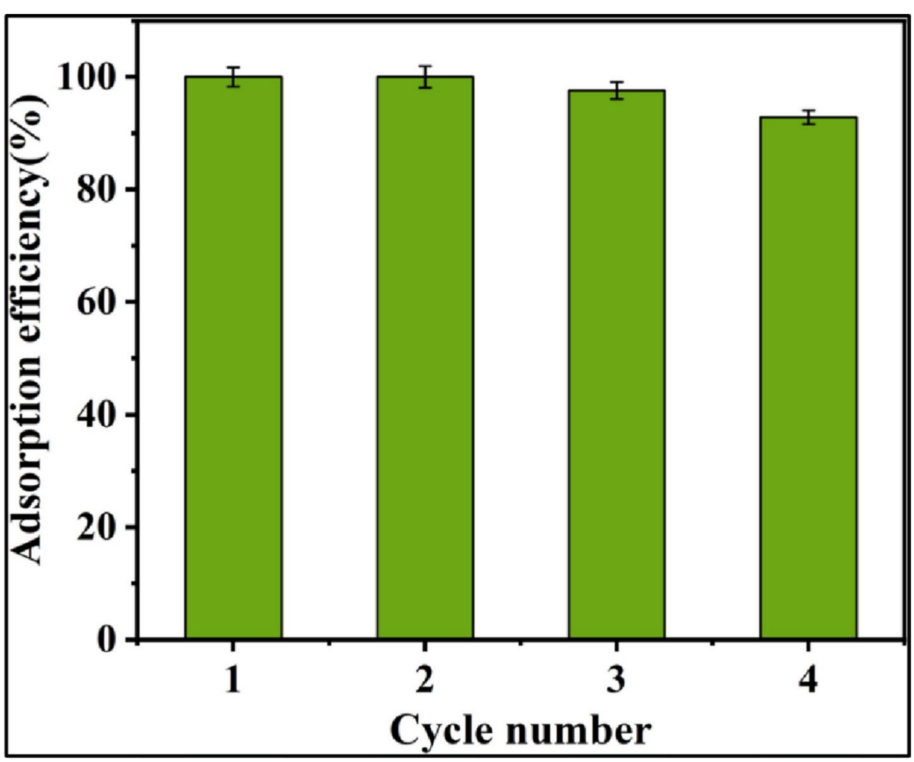


Fig. 17 The uranium adsorption efficiency of PAO-CS/GO2 in four successive reusing cycles [89]. Reproduced with the permission from [89]. Available under a CC-BY 4.0 license. Copyright 2018 SciEP

PAO-CS/GO2 beads have strong regeneration capabilities, which makes them ideal and eco-friendly for real-world, long-term uranium extraction from saltwater. These data collectively reveal that tri functionalization not only increases adsorption capacity but also improves selectivity, kinetics, and operational resilience [88, 89].

4.5.2 Advantages of tri functionalization

The capacity of tri functionalization to produce a multisite coordination environment that closely resembles the coordination preferences of uranyl ions is its main benefit. In comparison to simpler systems, the simultaneous presence of donor atoms based on nitrogen, oxygen, and phosphorus greatly improves binding strength and selectivity. Because of cooperative ligand effects that decrease desorption and inhibit competitive ion interference, tri functionalized GO has better adsorption capabilities. Improved adaptability to complicated aquatic environments, such as high ionic strength and changeable pH values, which are typical in actual nuclear effluent and natural waters, is another significant benefit. By promoting strong surface contacts and rapid diffusion, the multifunctional architecture further improves adsorption kinetics. Also, because the dispersed binding sites reduce localized stress and chemical degradation during repeated adsorption-desorption cycles, tri functionalization enhances reusability and long-term stability. Further flexibility for performance optimization for particular applications is provided by the tunability of functional group ratios [88, 89].

4.5.3 Limitations and challenges of tri functionalization

Despite its benefits, tri functionalization presents a number of drawbacks that need to be resolved before it can be used practically. One important restriction is synthetic complexity, since the coordinated insertion of three unique functional groups frequently requires multistep chemical modification methods, raising cost and lowering scalability. It is challenging to precisely manage the distribution of functional groups, and uneven functionalization might result in steric obstruction or decreased accessibility of active sites. The structural integrity of GO sheets may potentially be compromised by excessive ligand loading, which could have an impact on mechanical stability and dispersion. The possibility of inter ligand competition, in which several functional groups might compete for uranyl binding and, if improperly balanced, lower the effectiveness of cooperative interactions, is another difficulty. Also, even while regeneration efficiency is normally good, extended cycling may still cause functional groups to gradually disappear due to oxidative degradation or hydrolysis. These drawbacks emphasize the necessity of rigorous ligand composition management and optimal synthesis techniques [88, 89].

From Table 2., we can clearly say that a certain trend in adsorption performance is revealed by the comparison analysis: pure materials have limited capacity and selectivity, but functionalized and multifunctional systems perform better because of more active binding sites and synergistic interactions. However, the cost and complexity of synthesis could limit their practical use.

Table 2 Comparative analysis of functionalized carbon-based adsorbents for U(VI) extraction

Material	Functionalization	Capacity (mg/g)	Kinetics	Selectivity	Key Insights	Refs.
Activated carbon	Pristine	85	Pseudo-2nd	Low	Poor selectivity	[113]
GO-AC felt	Composite	131	Pseudo-2nd	Moderate	Hybrid improves adsorption	[40]
rGO/ZIF-67 aerogel	MOF hybrid	1888.5	Pseudo-2nd	Very High	Porous MOF boosts capacity	[28]
GO/alginate aerogel	Biopolymer	312	Pseudo-2nd	Moderate-High	3D network enhances uptake	[29]
Carbon Nanohorns	TETA-functionalized	256	Pseudo-2nd	High	Amine groups improve binding	[30]
CNT/chitosan aerogel	Composite	285	Pseudo-2nd	Moderate	Synergistic effect	[48]
Graphene oxide	Pristine	105	Pseudo-2nd	Low	Limited active sites	[56]
GO nanosheets	Pristine	97	Pseudo-2nd	Low	Weak adsorption	[57]
GO composite	Complexation	142	Pseudo-2nd	Moderate	Complexation improves selectivity	[58]
rGO/nZVI	Hybrid reduction	160	Pseudo-2nd	Moderate	Adsorption–reduction	[59]
GO/Polyamidoxime	Functionalized	320	Pseudo-2nd	High	More binding sites	[62]
rGO hydrogel	Hydrogel	210	Pseudo-2nd	Moderate	Porous aids diffusion	[63]
GO/amidoxime hydrogel	Amidoxime	298	Pseudo-2nd	High	Strong U(VI) chelation	[64]
Amidoxime polymer	Polymer	520	Pseudo-2nd	Very High	High binding density	[67]
Phosphate-GO	Phosphate	350	Pseudo-2nd	High	Phosphate enhances affinity	[68]
Magnetic phosphine oxide	Ligand-based	410	Pseudo-2nd	High	Ligand-driven selectivity	[69]
GO-chitosan beads	Phosphorylated	275	Pseudo-2nd	Moderate-High	Stable binding	[70]
Amine-GO composite	Amine	180	Pseudo-2nd	Moderate	Moderate improvement	[72]
Mn, N-doped carbon	Doped	220	Pseudo-2nd	Moderate-High	Electronic enhancement	[73]
N-doped porous carbon	COF-derived	265	Pseudo-2nd	Moderate-High	High surface area	[74]
Phosphate/amidoxime adsorbent	Dual ligand	430	Pseudo-2nd	High	Synergistic ligands	[77]
GO nanoribbon aerogel	Composite	340	Pseudo-2nd	High	Enhanced surface	[79]
GO-amidoxime-hydroxamic	Multi-ligand	610	Pseudo-2nd	Very High	Multiple coordination sites	[80]
ZnO/graphene composite	Hybrid	190	Pseudo-2nd	Moderate	Mixed adsorption	[83]

Table 2 (continued)

Material	Functionalization	Capacity (mg/g)	Kinetics	Selectivity	Key Insights	Refs.
GO-biopolymer gel	Amidoxime	295	Pseudo-2nd	High	Improved selectivity	[87]
Amidoxime chitosan beads	Functionalized	240	Pseudo-2nd	High	Strong uranium affinity	[88]
GO hydrogel beads	Amidoxime	305	Pseudo-2nd	High	Good regeneration	[88]
Mesoporous carbon	Functionalized	270	Pseudo-2nd	Moderate-High	High porosity	[99]
GO-modified adsorbent	Functionalized	150	Pseudo-2nd	Moderate	Better than pristine GO	[107]

5 Machine learning in uranium adsorbent design and optimization

ML was used to classify uranium adsorption performance and predict equilibrium adsorption capacity (Q_e) based on experimental factors including adsorbent type, pH, temperature, solid-liquid ratio, and initial uranium concentration. ML models can capture difficult, non-linear relationships between process variables and uranium uptake. Supervised learning models including k-nearest neighbour, Naive Bayes, logistic regression, support vector machines, convolutional neural networks (CNN), and random forest (RF), are used in this context. CNN was able to classify low and high adsorption levels with 98% accuracy in a recent uranium-specific study. RF showed the best Q_e regression performance (MSE = 334.45, R^2 = 0.92). Permutation-importance analysis revealed initial U concentration as the single most influential feature. These findings demonstrate how ML can quantitatively rank controlling parameters for adsorbent design and process by reducing the number of trial-and-error runs [114]. These supervised models directly learn from experimental adsorption datasets, enabling the prediction of Q_e for novel conditions and assisting in the design of cost-effective experiments [115]. In order to predict uranium (VI) sorption on mineral surfaces like quartz and montmorillonite, ML has been combined with geochemical speciation and surface complexation modelling in addition to batch adsorption testing. This is crucial for evaluating subsurface transport and repository safety. In order to avoid the need to calibrate conventional surface complexation models and still produce high-quality predictions for U(VI) adsorption driven by both ion-exchange and surface complexation reactions, a chemistry-informed hybrid approach (L-SURF) first computes aqueous speciation using PHREEQC and then feeds these speciation outputs as features to a random-forest model [116]. Similarly, for U sorption to oxides, deep neural network and random forest surrogates have been trained on over a million SCM-generated data points. These surrogates accurately replicate triple-layer model predictions at a fraction of the computational cost and without convergence issues, making it easier to incorporate sorption processes into large-scale radionuclide migration models for geological repositories [117].

Although many specific ML studies concentrate on other metals or organic pollutants, they reveal broad concepts that are extremely applicable to uranium extraction. Ten tree-based ML models (CatBoost, ExtraTrees, Boost, Random Forest, Light, and

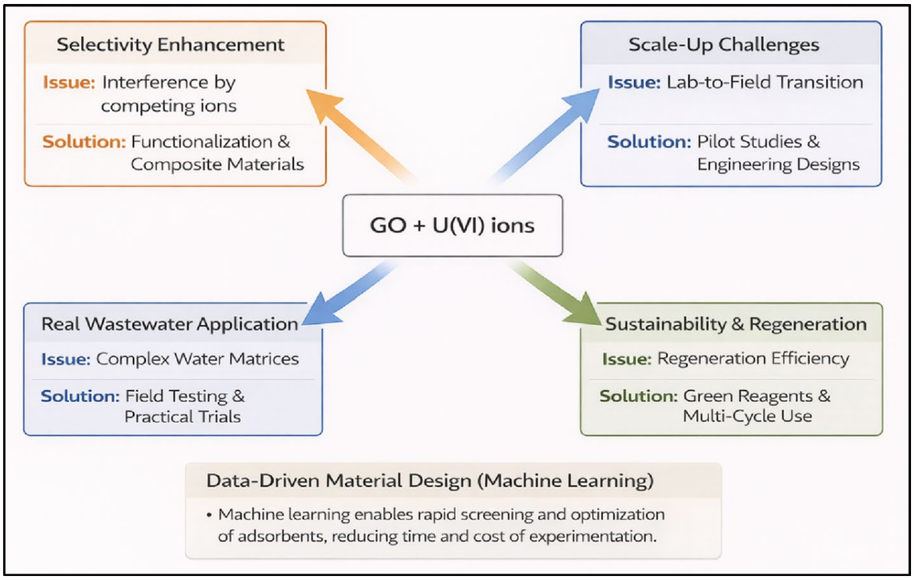


Fig. 18 Future perspectives in GO based uranium extraction

others) with 3,757 data points and 24 input variables were used in the work on emerging contaminants adsorption on biochar. The results showed that CatBoost achieved $R^2 \approx 0.94$ with low error. While SHAP analysis revealed that experimental conditions (such as initial concentration and contact time) contributed more to predictive power than synthesis conditions [115]. A pattern similar to the uranium study were initial concentration dominated feature importance. A study on arsenic (III) using nine predictors which describes adsorbent composition and experimental conditions was done by Mirza et al. The adsorption was compared with neural networks as well as non-neural network models (support vector machines, Gaussian process regression, linear regression, ensemble methods). Among all the models, GPR performed best with an R-squared value of 0.939, MAE of 5.778, and RMSE of 7.119 for training and an R-squared of 0.942, MAE of 5.450, and RMSE of 6.870 for testing [118]. All things considered, these adsorption-focused ML studies show a toolbox of models, such as CNN, RF, gradient-boosted trees, SVM, GPR, and deep neural networks, that can rank the importance of physicochemical and operating parameters, learn from large, noisy, multivariate datasets, and thus guide both adsorbent design and process optimization in uranium extraction systems.

6 Future perspectives

To address the constraints outlined in existing functionalization methodologies, future carbon-based uranium adsorbent development must increasingly incorporate coordination chemistry principles, materials engineering, and process scale considerations as shown in the Fig 18. Since carbonate competition and aqueous chemistry continue to determine adsorption performance in practical systems, a basic understanding of uranium speciation and surface complexation is still crucial. Therefore, in order to logically design ligands with optimal binding shape and stability under ecologically relevant conditions, advanced surface complexation modelling and spectroscopic investigations should be further utilized. Although the viability of uranyl capture was demonstrated by pristine and oxygen functionalized graphene materials, their low selectivity emphasizes the necessity of ligand directed architectures with controlled binding environments, especially within three-dimensional frameworks that improve mass transport and ligand accessibility. Future uranium recovery will still rely heavily on amidoxime-based systems, especially for seawater extraction. However, because of their vulnerability to oxidative degradation and carbonate interference, ligand stabilization through polymer backbone engineering, crosslinking techniques, and the addition of protective hydrophilic matrices is required. To extend their operational window across wider pH and salinity ranges, phosphoryl and phosphonate ligands which provide strong hard-hard coordination and chemical robustness should be further enhanced through regulated ligand spacing and hybridization with chelating groups. Future studies should concentrate on cooperative codoping and hybrid ligand design for amino and nitrogen doped materials in order to reduce oxidation effects and improve binding strength in acidic conditions. Multifunctional ligand systems provide an effective method for overcoming the limitations of single, dual, and tri-functional adsorbents by combining multiple binding sites that allow for stronger and more selective interactions with U(VI). Multifunctional materials can offer synergistic effects that enhance adsorption capacity and stability under challenging conditions, in comparison with simpler systems that frequently suffer from low selectivity and sensitivity to pH or competing ions. However, these systems also face a number of limitations, such as complex and expensive synthesis pathways, issues in achieving consistent functional group distribution, and possible steric hindrance or competitive interactions among ligands. Also, their practical application is challenged by high synthesis costs, limited scalability, and difficulties in large-scale manufacturing. Their practical application is made more difficult by problems with stability, regeneration, and a lack of knowledge on combined ligand effects.

Integrating ML will create a revolutionary impact on uranium extraction research in the future by facilitating data-driven adsorption system optimization and accelerating the creation of next-generation adsorbents. Using material parameters including surface chemistry, porosity, ligand density, pH levels, and competing ion environments, ML models can forecast uranium adsorption capacity, selectivity, and kinetic behavior rather than depending just on trial-and-error synthesis. ML can assist in determining the most important factors influencing uranyl binding and direct the logical design of high-performance adsorbents by learning complex structure property connections from experimental datasets. Also, thousands of potential materials may be computationally evaluated quickly with ML-assisted screening, greatly cutting down on the time and effort required for experiments. Advanced techniques such as random forest regression, neural networks, and inverse design frameworks can allow researchers to specify a desired adsorption performance and receive optimum material structures in return. Significantly, combining ML with high-throughput testing and molecular simulations may increase prediction accuracy for real-world systems like seawater, where multi-ion competition and carbonate complexation have a significant impact on adsorption. In order to overcome the gap between laboratory research and industrial application, ML-based predictive techniques will be crucial for designing adsorbents that balance capacity, stability, regeneration efficiency, and cost-effectiveness as uranium remediation advances toward scalable deployment. Even though ML is an innovative way to maximize uranium recovery, there are still a number of difficulties that need to be resolved before this method can be effectively used. First of all, it should be highlighted that there are significant gaps in the availability of adequate data sets for training reliable models in this area.

Also, it is difficult to assess the results and create models that are universally relevant when there is no standard technique for the research. It will be feasible to move from theory to practice if these issues are resolved.

7 Conclusion

The transition from single-functional to multifunctional ligand architectures in GO-based adsorbents has greatly increased uranium (VI) recovery by overcoming limitations in earlier designs. For uranyl binding, single-functional GO systems with oxygen groups, amino, amidoxime, or phosphoryl ligands rely on surface complexation, electrostatic attraction, and chelation. However, low active-site density, a limited pH operating range, poor resistance to carbonate and competing ions, and decreased selectivity in complicated nuclear waste and marine environments frequently limit their efficiency. Dual-functional GO materials combine various ligands, such as soft amidoxime groups with hard oxygen-donor phosphoryl/phosphate groups, to address these limitations. Adsorption capacity, speed, stability, and performance under various pH and ionic conditions are all enhanced by this. By enabling several binding methods simultaneously, including chelation, inner-sphere complexation, and electrostatic interactions, more sophisticated trifunctional GO systems offer even higher performance. Even in carbonate-rich and salty water, these systems exhibit extremely high uranium selectivity. They also exhibit improved mass transfer, more accessible active sites, and more mechanical strength when utilized in 3D hydrogel or aerogel structures, which makes them appropriate for actual wastewater and seawater uranium recovery. At the same time, ML is becoming a valuable tool in this industry. Without doing numerous trials, ML helps in the rapid testing and creation of new materials, the prediction of uranium adsorption performance, and the discovery of significant correlations between material structure and performance. Better uranium recovery methods can be developed more quickly by combining multifunctional GO materials with ML-based design and high-quality data. But there are still limitations with ML that restrict its accuracy and broad use, such as small datasets and inconsistent data reporting. A number of problems need to be fixed before laboratory research may move into real-world applications.

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